

Fraunhofer lines are dark lines in a light spectrum created by the absorption of certain wavelengths by gases. The pattern reveals the identity of the gas.

ON MARCH 13, 1813, BRITISH ENGINEER William Hedley (1779–1843) patented a design for a steam locomotive known as **Puffing Billy**. It began hauling coal trucks in Northumberland, England, in 1814 and is **the world's oldest surviving steam locomotive**. The greatest pioneer of steam railroads, **George Stephenson** (1781–1848), also

colors. When sunlight passed through the glass, Fraunhofer noticed **dark lines (Fraunhofer Lines) where color in the light spectrum was missing**. Fraunhofer was not the first to notice these lines, but in 1814 he was the first to start an extensive study of them, and in doing so provided a basis for the science of spectroscopy (see 1884–85).

“HE [WELLS] DISTINCTLY RECOGNIZES THE PRINCIPLE OF **NATURAL SELECTION**... THIS IS THE **FIRST RECOGNITION**... BUT HE APPLIES IT ONLY TO MAN.”

Charles Darwin, British naturalist, *The Origin of Species by Means of Natural Selection*, 4th edition, 1866

built his first steam locomotive in the north of England; it first ran on July 25, 1814.

In London, in 1813, American physician **William Wells** (1757–1817) read a paper to the Royal Society in which he explained racial differences on the basis of a **process of evolution involving natural selection**.

In Bavaria, German optician **Joseph von Fraunhofer** was making fine optical glass for separating light into different

Elsewhere in 1814, news was delivered with the help of **steam-powered presses** at *The Times* newspaper in London.

In Connecticut, inventor **Eli Terry** (1772–1852) developed a groundbreaking design for a **mass-produced clock** that could be made by machines instead of being hand-assembled by skilled clockmakers. This made clocks more affordable.

Mount Tambora on Sumbawa Island in Indonesia was the site the largest volcanic events in recorded history. The eruption took 4,593 ft (1,400 m) in height off the cone of the volcano.

MOUNT TAMBORA ON THE INDONESIAN ISLAND of Sumbawa erupted on April 5. It was the most **powerful volcanic eruption** in recorded history—the explosions could be heard as far away as 1,616 miles (2,600 km).

The role of fossils in studying Earth's history was revealed by British geologist **William Smith** (1769–1839).

Working as a surveyor, overseeing the digging of canals, Smith noticed that widely separated outcrops of the same **rock strata could be identified by the fossils they contain**. He used this to create the first geological map in 1799, and in 1815 he published a geological map of Britain. His map

became the model for all geological maps.

The canals Smith helped to build were essential to the accelerating Industrial Revolution in Britain, as were the mines that provided coal for fires and steam engines. Mining was dangerous work, however, and miners lived in constant fear of hitting pockets of methane or other flammable gases, known as firedamp, which could explode if they reached the

naked flame of their candles. British scientist **Humphry Davy** invented the **miners' safety lamp**. The lamp's flame was wrapped in wire mesh that reduced the chance of it igniting gases.

In chemistry, the atomic theory of elements was gaining supporters. British scientist **William Prout** (1785–1850) concluded from studying tables of atomic weights (see 1803–04) that every weight is a multiple of the weight of a hydrogen atom, and that the **hydrogen atom is the only fundamental particle** from which all other elements are made up. He was not right, but a century later, in 1920,



William Smith's geological map
This pioneering map of Britain showed the geological make-up of the country and set a precedent for geological maps in the future.

Ernest Rutherford (1871–1937) named the proton partly in honour of Prout.

In France, scientist **Jean-Baptiste Biot** (1774–1862) was **experimenting with polarized light**—light vibrating in just a single plane (see panel, opposite). On October 23, he shone a beam of polarized light through a tube of turpentine and noticed how the plane of polarization was rotated. Other organic liquids, such as lemon juice, produced the same effect. This rotation is at the heart of the Liquid Crystal Displays (LCD) now widely used in display screens.

wire mesh prevents flame from igniting mine gasses

Davy lamp

This miner's safety lamp consisted of a cylinder of wire gauze containing a wick attached to an oil reservoir.



March 13, 1813 British-American physician William Wells makes the first clear statement about **evolution by natural selection**

July 25, 1814 British engineer George Stephenson introduces his first **steam locomotive**, Blucher

1814 Joseph von Fraunhofer creates the science of **spectroscopy** and discovers **Fraunhofer lines**

November 29, 1814 The London Times newspaper is **printed by steam-driven press** for the first time

1814 William Hedley's **Puffing Billy** steam locomotive goes into service

1814 American inventor Eli Terry creates the **first mass-produced clock**

April 5 Mount Tambora erupts in Indonesia

William Prout suggests the **hydrogen atom** is the only fundamental particle

William Smith publishes the **first geological map** of Britain

November 9 Humphry Davy presents the **miners' safety lamp** to the Royal Society

Jean-Baptiste Biot shows how **polarized light** is rotated by liquids